

UQFF Advanced Integration: Wormhole-MUGE | Meissner Exponential Relativistic γ | Error Propagation

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Abstract

This paper presents four new mathematical formulations extending the UQFF framework: (1) a wormhole-MUGE coupling term from the Morris-Thorne metric; (2) the exponential Meissner superconductivity model replacing the linear B/B_{crit} quenching (PAPER_372); (3) a special-relativistic Lorentz correction applied to the DPM acceleration for high-velocity systems; and (4) a formal error propagation formalism for uncertainty quantification. All four are combined into the Unified UQFF master equation and validated for J1610+1811 at $v=0.99c$.

1. Wormhole-MUGE Coupling Term

The Morris-Thorne wormhole geometry (PAPER_373) introduces a new acceleration term:

$$a_{\text{worm}}(r) = f_{\text{worm}} * E_{\text{vac,neb}} / (b^2 + r^2)$$

where $f_{\text{worm}} = 10^{-10}$ (dimensionless coupling), $E_{\text{vac,neb}} = 7.09 \times 10^{-36}$ J, $b = 1.0$ m (throat). This encodes the gravitational contribution of a wormhole throat modulated by local vacuum energy density.

At $r = b = 1$ m:

$$a_{\text{worm}}(1) = 1e-10 * 7.09e-36 / (1 + 1) = 3.545e-46 \text{ m/s}^2$$

2. Meissner Exponential Superconductivity

PAPER_372 uses the linear Meissner approximation $(1 - B/B_{\text{crit}})$. This paper introduces the physically more accurate **exponential form** applicable to Type-II superconductors (London penetration depth model):

$$\text{Linear: } (1 - B/B_{\text{crit}})$$

$$\text{Exponential: } \exp(-B/B_{\text{crit}})$$

The exponential form ensures monotone decay from $e^0 = 1.0$ (no field) to 0 (field well above B_{crit}), without the unphysical negative values that arise from the linear form when $B > B_{\text{crit}}$.

System	B/B_{crit}	Linear factor	Exponential factor
SGR1745-2900	0.1	0.9	0.905

System	B/B_crit	Linear factor	Exponential factor
Sagittarius A*	0.1	0.9	0.905
Student's Guide	0.1	0.9	0.905

Table 1. Linear vs exponential Meissner factor comparison

3. Relativistic Lorentz Correction

For high-velocity systems (J1610+1811 jet at $v = 0.99c$), the DPM acceleration undergoes Lorentz suppression:

$$\gamma = 1 / \sqrt{1 - v^2/c^2}$$

$$a_{\text{DPM}} \rightarrow a_{\text{DPM}} / \gamma$$

For $v = 0.99c$:

$$\gamma = 1 / \sqrt{1 - 0.9801} = 1 / \sqrt{0.0199} = 7.089$$

This suppresses a_{DPM} by factor ~ 7.09 , reflecting that the DPM force (electromagnetic in origin) is frame-dependent in the relativistic regime. All other resonance terms retain their coordinate-frame values.

4. Error Propagation Formalism

Uncertainties in individual MUGE terms propagate in quadrature:

$$\delta a_g = \sqrt{\sum_i (\delta a_i)^2}$$

For a fractional error $f = 1\%$:

$$\delta a_i = f * |a_i| \rightarrow \delta a_g = f * \sqrt{\sum_i a_i^2}$$

This provides a rigorous uncertainty bound enabling comparison with observational error bars.

5. Unified UQFF Master Equation (Complete Form)

Combining PAPER_371–375:

[Compressed UQFF – PAPER_372, Meissner exp]:

$$GM/r^2 * (1+H_0*t) * \exp(-B/B_{\text{crit}}) * (1+F_{\text{env}})$$

$$+ U_g_{\text{sum}} + \Lambda c^2/3 + \text{quantum_integral} + \text{fluid} + \text{perturbation}$$

[MUGE Resonance – PAPER_371, Lorentz corrected]:

$$+ a_{\text{DPM}}/\gamma + a_{\text{THz}} + a_{\text{vac,diff}} + a_{\text{super,freq}} + a_{\text{aether,res}} + U_{g4i}$$

$$+ a_{\text{quantum,freq}} + a_{\text{Aether,freq}} + a_{\text{fluid,freq}} + a_{\text{osc}} + a_{\text{exp,freq}} + f_{\text{TRZ}}$$

[Wormhole – PAPER_373]:

$$+ a_{\text{worm}} = f_{\text{worm}} * E_{\text{vac,neb}} / (b^2 + r^2)$$

[Total uncertainty – PAPER_375]:

$$\pm \delta a_g = \sqrt{\sum_i (\delta a_i)^2}$$

6. Canonical Parameter Summary

Symbol	Value	Source Paper
f_worm	1e-10	PAPER_375
Meissner form	$\exp(-B/B_{\text{crit}})$	PAPER_375 (vs linear PAPER_372)
gamma (v=0.99c)	~7.089	PAPER_375
delta_g (1% error, SGR1745)	~1e-11	PAPER_375

Table 2. New parameters introduced in PAPER_375

7. Implementation

C++: STAR_MAGIC_09SEPT_UQFF_MODULE.cpp, namespace UQFFAdvanced

- compute_a_wormhole(Evac_neb, b, r)
- meissner_exp(B, Bcrit)
- lorentz_gamma(v)
- apply_lorentz(aDPM, v)
- error_propagation(delta_terms)
- compute_unified_UQFF(sys, res, t, v_jet, b_worm, r_worm)
- compute_total_uncertainty(sys, p, frac_error)

Python: CondensedPhysics4.py, class
UQFFWormholeMeissnerRelativisticGammaCalculator (CP4 #23)

Wolfram macro: WOLFRAM_TERM_UQFF_ADVANCED